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CSA1717 – ARTIFICIAL INTELLIGENCE

CO1 – LIST LAB PROGRAMS PYTHON

EXP 9: Travelling Salesman Problem

PROGRAM:

practical lab 9.py - C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 9.py (3.14.5)

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```
from itertools import permutations

cities = ["A", "B", "C", "D"]

distance = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

best_cost = float("inf")
best_route = None

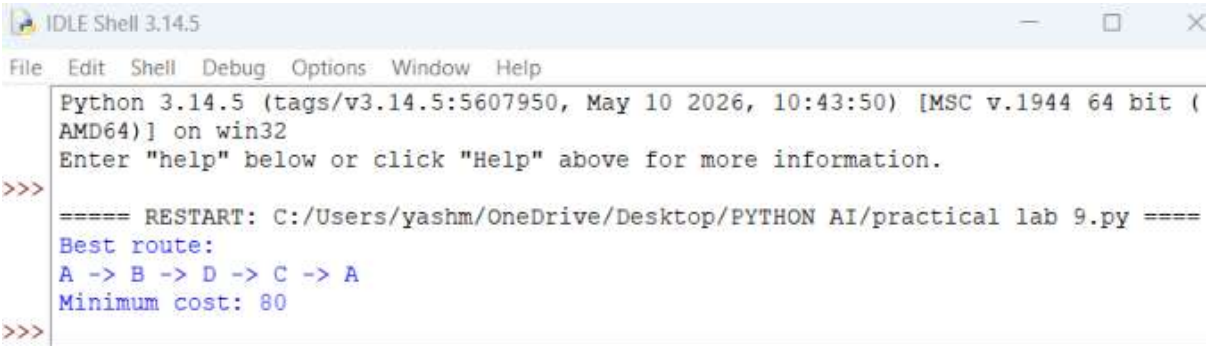
for order in permutations([1, 2, 3]):
    route = [0] + list(order) + [0]
    cost = 0

    for i in range(len(route) - 1):
        cost += distance[route[i]][route[i + 1]]

    if cost < best_cost:
        best_cost = cost
        best_route = route

print("Best route:")
print(" -> ".join(cities[i] for i in best_route))
print("Minimum cost:", best_cost)
```

OUTPUT:


IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 9.py ====
Best route:
A -> B -> D -> C -> A
Minimum cost: 80
>>>